

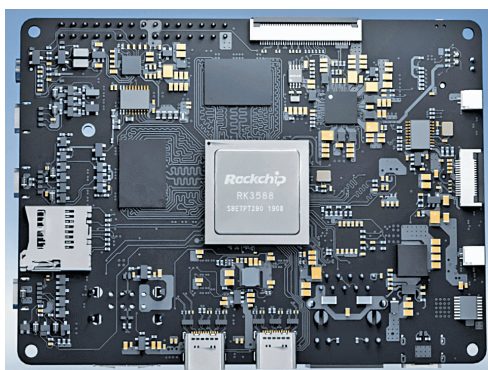
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1 VICHARAK'S DEVELOPMENT BOARDS THAT FILL THE GAPS LEFT BY RASPBERRY PI AND OTHER BOARDS

This Indian startup is making edge AI hardware more accessible, equipping developers to build faster, smarter machines without high costs or reliance on the cloud.



Axon Board



Vaaman Board

Building hardware is often difficult for people with strong ideas but limited resources. Tools are expensive, ecosystems are fragmented, and turning concepts into real products can feel out of reach. Akshar Vastarpara, coming from a software background, saw that innovation was slowed less by talent and more by a lack of access. He believed hardware should be as approachable as software and founded Vicharak to make it easier, faster, and more accessible to build. 'Vicharak,' meaning someone who thinks, reflects the company's vision of computation as a tool to think, process, and create.

Vicharak builds edge-computing hardware for AI applications, focusing on single-board computers comparable to Raspberry Pi and

Nvidia Jetson platforms, but with built-in AI acceleration and stronger developer support. The goal is to enable physical AI systems, such as robots, drones, and industrial machines, to process data on-device rather than relying on the cloud, thereby improving speed and efficiency. Its main board, Axon, is an all-in-one platform with a processor, memory, storage, connectivity, and motor-control pins, allowing developers to directly connect sensors and build applications.

While Raspberry Pi is widely used, modern edge AI applications demand more compute power. Akshar explains, "While Raspberry Pi revolutionised the market by focusing on hobbyists, education, and lightweight edge computing, its hardware still

comes with limitations for modern AI-heavy workloads. Our Axon board addresses this gap with higher-performance processors, dedicated AI acceleration, support for multiple camera streams, richer storage, and expanded sensor and display interfaces for more demanding edge applications."

Beyond Axon, Vicharak launched the Shrike-lite board, a unique fusion of an Arduino-style development experience with FPGA capabilities. Designed for beginners, students, and early-stage engineers, Shrike-lite lowers the barrier to entering electronics, computation, and chip design.

Another major product line is Vaaman. Unlike traditional single-board computers, Vaaman combines a Linux-capable compute platform with FPGA technology in a single compact form factor, enabling users to experiment with much higher-performance, hardware-accelerated computing than standard boards can offer.

Talking about the design challenges faced in developing these boards, Akshar adds, "When we started building early platforms like Vaaman with a 10-layer PCB and later moved to Axon with a 12-layer design, we were stepping into a level of complexity we didn't fully understand yet. As we pushed towards 14-layer, 16-layer, and eventually HDI-class boards, the biggest